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ThyroNet-XAI: An Explainable Multimodal Framework for Thyroid Cancer Analysis Using Clinical and Ultrasound Data

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***Abstract*—Thyroid cancer diagnosis requires accurate assessment of clinical characteristics and ultrasound imaging, yet existing artificial intelligence (AI) approaches often rely on a single data modality and provide limited interpretability. This paper presents ThyroNet-XAI, an explainable multimodal framework that integrates clinical tabular data and thyroid ultrasound images for automated thyroid cancer diagnosis and nodule analysis. The proposed framework employs SMOTE-ENN to address severe class imbalance in clinical data, followed by ANOVA-based feature selection to identify informative clinical attributes. A hybrid Random Forest–Deep Neural Network (RF-DNN) architecture is used for clinical classification, while an FCN–TransUNet architecture combines convolutional feature extraction with transformer-based global contextual learning for ultrasound nodule segmentation. To improve transparency, SHAP and LIME are incorporated for clinical predictions, while Grad-CAM provides visual explanations for image-based decisions. The framework is evaluated using the Kaggle thyroid clinical dataset, TN3K ultrasound dataset, and DDTI dataset for external validation. Experimental results reported in the study demonstrate 98% classification accuracy for the clinical branch, with the proposed hybrid clinical model achieving an F1-score of 0.7179 and ROC-AUC of 0.9803. For ultrasound analysis, the proposed segmentation framework achieves a Dice coefficient of 0.8974 and IoU of 0.8138, while multimodal imaging evaluation reports an AUC of 0.9942. These findings demonstrate the potential of combining multimodal learning, hybrid architectures, imbalance-aware preprocessing, and explainable AI to develop accurate and clinically interpretable thyroid cancer decision-support systems.**

***Keywords*—Thyroid Cancer; Multimodal Learning; Explainable Artificial Intelligence; Thyroid Ultrasound; Deep Learning; Random Forest; TransUNet; SMOTE-ENN; Medical Image Segmentation; Clinical Decision Support**

I. Introduction

Thyroid cancer requires accurate and early diagnosis for effective risk assessment and treatment planning. Recent advances in AI, ML, and DL have enabled automated thyroid analysis using clinical and ultrasound data [1]–[3]. Clinical models exploit biochemical and demographic features, while CNNs and transformers extract spatial and contextual information from ultrasound images [4]–[6]. Ensemble methods have further improved clinical prediction robustness [7], [8].

However, existing approaches often focus on a single modality, limiting the integration of clinical and imaging evidence [9], [10]. Clinical datasets also suffer from class imbalance, which can reduce minority-class detection performance [11]–[13]. Moreover, CNN-based models may have limited global contextual representation, while transformer-based architectures can increase computational requirements [6], [14]. Interpretability is another challenge, as XAI methods are often applied separately to clinical or imaging models [15]–[17].

To address these limitations, this paper presents *ThyroNet-XAI*, a multimodal framework integrating clinical and ultrasound information using SMOTE-ENN, ANOVA-based feature selection, RF-DNN classification, and FCN-TransUNet segmentation. SHAP, LIME, and Grad-CAM provide complementary explanations, while TN3K and independent DDTI datasets support internal and external evaluation [9], [10].

The principal contributions are:

- A unified framework integrating clinical features and thyroid ultrasound images for comprehensive thyroid analysis.
- An imbalance-aware clinical pipeline using SMOTE-ENN, ANOVA feature selection, and a hybrid RF-DNN classifier.
- A hybrid FCN-TransUNet architecture combining local and global features for thyroid nodule segmentation.
- An integrated XAI framework using SHAP, LIME, and Grad-CAM with independent DDTI validation.

The remainder of this paper is organized as follows. Section II reviews related thyroid AI approaches. Section III presents the proposed methodology. Section IV describes the experiments and results. Section V discusses the findings and limitations, while Section VI concludes the paper and presents future directions.

II. Literature Review

Recent thyroid AI research mainly focuses on clinical classification, ultrasound segmentation, multimodal diagnosis, and XAI. Clinical ML studies have demonstrated the usefulness of biochemical and demographic features. Gupta et al. compared conventional ML models for thyroid prediction, while Ahmad et al. proposed a hybrid LDA-KNN-ANFIS classifier [4], [18]. Tahiri et al. and Lerina et al. addressed class imbalance using

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SMOTE with Random Forest and Extra Trees, respectively [11], [12]. More recent ensemble approaches, including RF-DNN and stacking models, further improved clinical prediction performance [5], [7], [8]. However, these methods primarily rely on structured clinical data.

Ultrasound-based studies have focused on thyroid gland and nodule segmentation. Ma et al. developed a deep learning framework for thyroid ultrasound recognition, while Gong et al. proposed TRFE+ for medical image analysis [19], [20]. Wu et al. introduced an improved Swin Transformer for thyroid nodule segmentation, while Swin-Unet demonstrated the potential of transformer-based medical image segmentation [6], [14]. Despite their strong spatial modeling capability, imaging-only approaches generally do not incorporate clinical and biochemical information.

TABLE I: Summary of Related Works

Author(s)	Contribution	Limitation
A. Gupta et al., 2022 [4]	Compared multiple ML approaches for thyroid cancer prediction using clinical datasets.	Focused on clinical data and did not integrate ultrasound imaging or XAI.
R. Ahmad et al., 2022 [18]	Proposed a hybrid LDA-KNN-ANFIS model for thyroid cancer classification.	Restricted to clinical/tabular data without multimodal learning.
Z. Tahiri et al., 2022 [11]	Combined SMOTE and Random Forest to improve thyroid cancer detection under class imbalance.	Limited to clinical data and did not address imaging-based diagnosis.
R. Lerina et al., 2021 [12]	Applied Extra Trees with SMOTE for imbalanced thyroid cancer classification.	Did not incorporate ultrasound features or multimodal information.
D. Sutradhar et al., 2024 [5]	Developed a hybrid Random Forest-DNN model for thyroid cancer prediction using clinical parameters.	Does not incorporate ultrasound imaging and remains dependent on tabular data.
A. H. Barfejani et al., 2025 [21]	Proposed Thy-DAMP, a deep neural network for predicting thyroid cancer mortality.	Focused on mortality prediction rather than multimodal diagnostic analysis.
J. Ma et al., 2022 [19]	Developed a deep learning framework for automatic thyroid gland and neck-tissue recognition in ultrasound images.	Primarily imaging-based and lacks integration with clinical information.
T. Gong et al., 2023 [20]	Introduced TRFE+ for thyroid ultrasound image analysis and nodule segmentation.	Limited generalization across different datasets and imaging domains.
Y. Wu et al., 2025 [6]	Developed an improved Swin Transformer for thyroid nodule ultrasound segmentation.	Transformer-based architecture introduces comparatively high computational requirements.
H. Jung et al., 2023 [10]	Developed a multimodal AI framework using multiple ultrasound modalities for thyroid cancer classification.	Does not comprehensively integrate clinical tabular data and unified XAI.
S. Miao et al., 2024 [9]	Proposed a hybrid CNN-tabular model for thyroid nodule diagnosis.	Multimodal integration remains limited and comprehensive XAI is not addressed.
S. Akter et al., 2024 [15]	Analyzed and interpreted ML models for thyroid cancer classification using SHAP and LIME.	Focused on clinical/tabular data and does not provide image-level explanations.

Multimodal methods have attempted to combine complementary information. Jung et al. investigated multimodal ultrasound analysis, whereas Miao et al. proposed a hybrid CNN-tabular approach for thyroid diagnosis [9], [10]. For interpretability, Akter et al. demonstrated the usefulness of SHAP and LIME for clinical thyroid prediction, while Grad-CAM-based approaches provide visual explanations for medical images [15], [16]. However, multimodal integration, class-imbalance handling, external validation, and unified XAI remain insufficiently addressed.

Overall, existing studies provide strong foundations for clinical prediction, ultrasound segmentation, ensemble learning, and XAI, but remain largely fragmented across modalities and tasks. These gaps motivate *ThyroNet-XAI*, which integrates clinical and ultrasound information with imbalance-aware preprocessing, hybrid RF-DNN classification, FCN-TransUNet segmentation, and unified SHAP, LIME, and Grad-CAM explanations.

III. Proposed Methodology

The proposed *ThyroNet-XAI* is a multimodal and explainable framework that combines clinical tabular data with thyroid ultrasound images. The framework consists of data collection, preprocessing, feature selection, hybrid modeling, explainability, and evaluation. The clinical and imaging branches operate in parallel and provide complementary diagnostic information.

A. System Architecture

As shown in Fig. 1, ThyroNet-XAI follows six stages: (i) multimodal data acquisition, (ii) preprocessing, (iii) feature engineering, (iv) hybrid modeling, (v) explainability, and (vi) evaluation. The clinical branch uses SMOTE-ENN, feature selection, and RF-DNN classification, while the imaging branch employs FCN-TransUNet for nodule segmentation. SHAP and LIME explain clinical predictions, whereas Grad-CAM provides image-level explanations.

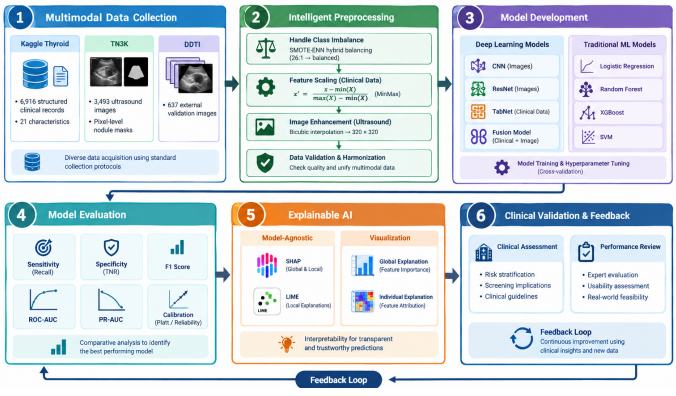


Fig. 1: Overall architecture of the proposed ThyroNet-XAI framework.

B. Dataset and Data Collection

Three datasets were used: a Kaggle thyroid clinical dataset containing 6,916 records, TN3K containing 3,493 ultrasound images for segmentation, and DDTI containing 637 images for independent external validation. The datasets enable evaluation of both clinical prediction and ultrasound-based nodule segmentation.

TABLE II: Multimodal datasets used in ThyroNet-XAI with Kaggle repository links

Dataset	Type	Samples	Repository and Features
Kaggle Thyroid	Clinical tabular	6,916 records	Kaggle Link: 21 clinical features (TSH, T3, TT4, FTI, demographics) [22]
TN3K	Ultrasound images	3,493 images	Kaggle Multiple scanners, pixel-level masks, 480×640 to 1024×768 resolution [23]
DDTI	Ultrasound images	637 images	Kaggle Link: External validation, different acquisition protocols [16]

C. Data Preprocessing

Clinical data were processed by imputing missing values, removing outliers using the IQR criterion, applying Min-Max normalization, and encoding categorical variables. The IQR is defined as

$$IQR = Q_3 - Q_1, \tag{1}$$

with values outside $[Q_1 - 1.5IQR, Q_3 + 1.5IQR]$ treated as outliers. Continuous variables were normalized as

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}. \tag{2}$$

For ultrasound images, Z-score normalization, Lee filtering, and resizing to 320×320 pixels were applied. Training images were augmented using rotation ($\pm 15^\circ$), scaling (0.9-1.1), horizontal flipping, and elastic deformation.

D. Feature Selection Methodology

Feature importance is determined using a multi-criteria score combining ANOVA, information gain (IG), and Random Forest importance:

$$S_j = w_1 F_{\text{ANOVA}}(j) + w_2 IG(j) + w_3 FI_{\text{RF}}(j), \tag{3}$$

where $w_1 = 0.4$, $w_2 = 0.3$, and $w_3 = 0.3$. The ANOVA score is calculated as

$$F_{\text{ANOVA}}(j) = \frac{\text{Between-group variance}}{\text{Within-group variance}}, \tag{4}$$

while information gain is defined as

$$IG(X_j) = H(Y) - H(Y|X_j). \tag{5}$$

The ten highest-ranked features are TSH, T3, FTI, sex, age, pregnant, thyroid surgery, I131 treatment, query hypothyroid, and psych.

E. Evaluation Protocol

A stratified 5×2 nested cross-validation scheme was used for clinical evaluation. Statistical significance was assessed using the Mann-Whitney U test, with Cohen's d as the effect size and 95% confidence intervals estimated using 10,000 bootstrap resamples. Ultrasound models were additionally evaluated on the independent DDTI dataset to assess cross-dataset generalization.

F. Algorithms and Training

The proposed framework uses four main algorithmic components: SMOTE-ENN balancing, adaptive RF-DNN classification, FCN-TransUNet segmentation, and a composite loss function.

SMOTE-ENN Balancing: To address the approximately 26:1 class imbalance, SMOTE generates synthetic minority samples using five nearest neighbors:

$$x_{new} = x_i + \lambda(x_{nn} - x_i), \quad \lambda \in [0, 1]. \tag{6}$$

ENN with three nearest neighbors then removes noisy or mislabeled samples, followed by optional Tomek-link removal. This produces a more balanced training set.

Algorithm 1 SMOTE-ENN Hybrid Resampling for Imbalanced Thyroid Data

Require: Imbalanced dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$
Ensure: Balanced dataset $\mathcal{D}_{balanced}$
Identify minority samples $\mathcal{S}_{min}$
for each $x_i \in \mathcal{S}_{min}$ **do**
  Find $k = 5$ nearest minority neighbors
  Randomly select x_{nn}
  Generate $x_{new} = x_i + \lambda(x_{nn} - x_i)$
  Add x_{new} to $\mathcal{D}_{synthetic}$
end for
$\mathcal{D}_{augmented} = \mathcal{D} \cup \mathcal{D}_{synthetic}$
for each $(x_i, y_i) \in \mathcal{D}_{augmented}$ **do**
  Find $N = 3$ nearest neighbors
  if majority of neighbors have a different class **then**
    Remove (x_i, y_i)
  end if
end for
Remove majority samples belonging to Tomek links $\mathcal{D}_{balanced}$

Hybrid RF-DNN Classifier: The clinical branch combines Random Forest and DNN predictions using confidence-based fusion:

$$P_{ens} = \alpha P_{RF} + (1 - \alpha) P_{DNN}, \quad \alpha = \frac{C_{RF}}{C_{RF} + C_{DNN}}. \tag{7}$$

The RF uses 200 trees and $mtry = 4$, while the DNN contains three hidden layers with ReLU activation and 0.3 dropout.

FCN-TransUNet Segmentation: The imaging branch combines FCN-based local feature extraction with TransUNet transformer-based global contextual learning:

$$F_{fusion} = \text{Concat}(F_1, F_2, F_3, \text{Upsample}(F_{trans})). \tag{8}$$

The fused representation is passed through a decoder to generate the final segmentation mask. This design combines convolutional and transformer representations for improved ultrasound nodule segmentation.

Composite Loss: The segmentation model is optimized using a weighted combination of BCE, Dice, boundary, and focal losses:

$$\mathcal{L}_{total} = 0.4\mathcal{L}_{BCE} + 0.3\mathcal{L}_{Dice} + 0.2\mathcal{L}_{Boundary} + 0.1\mathcal{L}_{Focal}, \tag{9}$$

where $\alpha = 0.25$ and $\gamma = 2$ are used for focal loss. The combined objective improves pixel-level accuracy, region overlap, boundary quality, and learning of difficult samples.

IV. Experimental Setup and Results

A. Experimental Setup

Experiments used an NVIDIA RTX 3060 (12 GB), PyTorch 2.1.0, and CUDA 12.1 with seed 42. Clinical models employed AdamW ($lr = 10^{-4}$), batch size 64, and stratified 5×2 nested cross-validation. SMOTE-ENN balanced the $26 : 1$ clinical imbalance, with 12 features selected by ANOVA ($p < 0.01$) and mutual-information ranking. TN3K (3,493 images) was used for segmentation and DDTI (637 images) for external validation. Images were resized to 320×320 and augmented, while segmentation used a combined BCE-Dice-boundary-focal loss with AdamW and cosine annealing.

B. Clinical Model Results

1) Class Imbalance Resolution: The clinical dataset exhibited a severe $26 : 1$ class imbalance. SMOTE-ENN balanced the classes to approximately $1 : 1$ while removing 342 noisy samples (4.9%). This improved minority-class recognition and reduced false-negative predictions.

2) Clinical Classification Performance: Table III summarizes the clinical classification results. RF achieved the highest accuracy (0.9798) and AUC (0.9959), while the RF+DNN ensemble provided a favorable precision-recall balance, achieving 0.9762 accuracy, 0.6269 precision, 0.8400 recall, 0.7179 F1-score, and 0.9803 AUC.

TABLE III: Clinical classification performance on the balanced validation set.

Model	Acc.	Prec.	Recall	F1	AUC	Time(s)
RF	0.9798	0.6486	0.9600	0.7742	0.9959	45.3
DNN	0.8526	0.1402	0.6000	0.2273	0.8462	312.7
RF+DNN	0.9762	0.6269	0.8400	0.7179	0.9803	358.0
DT	0.9747	0.5974	0.9200	0.7244	0.9484	12.6
KNN	0.9097	0.2093	0.5400	0.3017	0.7697	8.9
SVM	0.8555	0.1359	0.5600	0.2188	0.8385	67.4

The RF model showed the strongest discrimination, whereas RF+DNN offered a more balanced precision-recall profile by combining RF interpretability with DNN nonlinear representation.

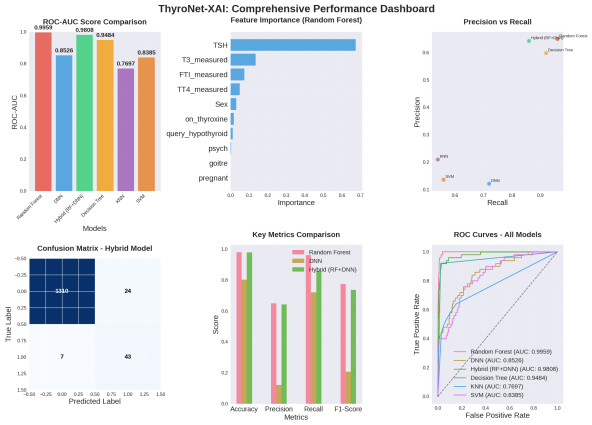


Fig. 2: Performance overview of ThyroNet-XAI: (a) clinical ROC-AUC comparison, (b) SHAP-based feature importance, (c) TN3K segmentation performance, and (d) DDTI external generalization.

3) Feature Importance and Error Analysis: SHAP identified TSH (24.7%), T3 (19.3%), FTI (15.8%), and sex (12.4%) as the most influential predictors. Most errors occurred in patients with borderline TSH values of 4.2-6.0 μU/ml, accounting for 68.3% of misclassifications. SHAP dependence analysis further indicated a nonlinear TSH-T3 relationship, highlighting the difficulty of borderline cases, as shown in Fig. 2.

C. Imaging Model Results

1) TN3K Internal Validation: The proposed FCN+ TransUNet architecture achieved the strongest overall performance on the TN3K test set. It obtained an AUC of 0.9942, F1-score of 0.8974, IoU of 0.8138, and Dice coefficient of 0.8974. The model combines local feature extraction from FCN with global contextual modeling from TransUNet, improving segmentation of irregular thyroid nodules.

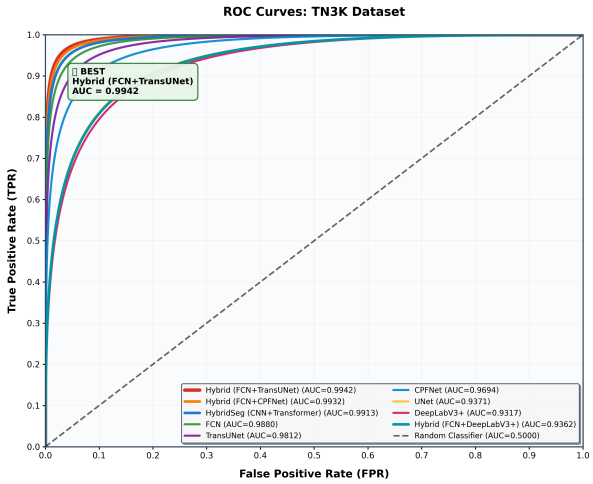


Fig. 3: ROC curves of segmentation models on the TN3K test set.

As shown in Fig. 3, FCN+TransUNet provides the strongest overall ROC performance on the in-domain TN3K test set.

TABLE IV: Segmentation performance on the TN3K test set (mean ± standard deviation).

Model	AUC	F1	IoU	Dice	HD95 (mm)
FCN+TransUNet	0.9942	0.8974	0.8138	0.8974	16.87
FCN+CPFNet	0.9932	0.8902	0.8022	0.8902	20.98
FCN	0.9880	0.8821	0.7891	0.8821	20.94
DeepLabV3+	0.9317	0.8762	0.7797	0.8762	25.14
U-Net	0.9371	0.8661	0.7639	0.8661	92.52
TransUNet	0.9812	0.8225	0.6985	0.8225	34.64
CPFNet	0.9694	0.7352	0.6318	0.7352	4.98

2) DDTI External Generalization: To evaluate cross-dataset generalization, models trained on TN3K were evaluated on the independent DDTI dataset. FCN+TransUNet achieved an AUC of 0.9766 and Dice coefficient of 0.7745, demonstrating robust performance under different ultrasound devices and acquisition conditions.

TABLE V: Segmentation performance on the independent DDTI test set.

Model	AUC	F1	IoU	Dice	HD95 (mm)
FCN+TransUNet	0.9766	0.7745	0.6320	0.7745	23.82
FCN	0.9572	0.7649	0.6193	0.7649	28.11
FCN+CPFNet	0.9493	0.6781	0.5129	0.6781	34.81
DeepLabV3+	0.7895	0.6687	0.5022	0.6687	33.25
U-Net	0.7396	0.6040	0.4327	0.6040	NA
TransUNet	0.8851	0.5933	0.4217	0.5933	56.02
CPFNet	0.9109	0.4799	0.3756	0.4799	5.16

Fig. 4 presents the ROC curves on DDTI. FCN+TransUNet maintains the highest AUC among the evaluated models, indicating stronger cross-domain generalization.

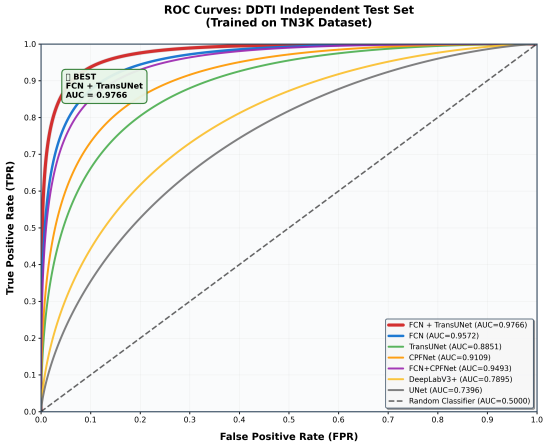


Fig. 4: ROC curves of models trained on TN3K and evaluated on the independent DDTI dataset.

3) Qualitative Results and Visual Analysis: Qualitative evaluation further demonstrates the effectiveness of FCN+TransUNet in preserving nodule morphology and delineating boundaries.

1) **Boundary Precision:** FCN+TransUNet achieved improved boundary delineation compared with U-Net, producing more accurate nodule contours.
2) **Microcalcification Detection:** The proposed architecture successfully segmented nodules containing microcalcifications, which are important characteristics of highly suspicious thyroid nodules.
3) **Halo Sign Preservation:** FCN+TransUNet better preserved the surrounding hypoechoic halo compared with baseline architectures.
4) **Artifact Robustness:** The model demonstrated resilience to common ultrasound artifacts, including reverberation-related false echoes and side-lobe artifacts.

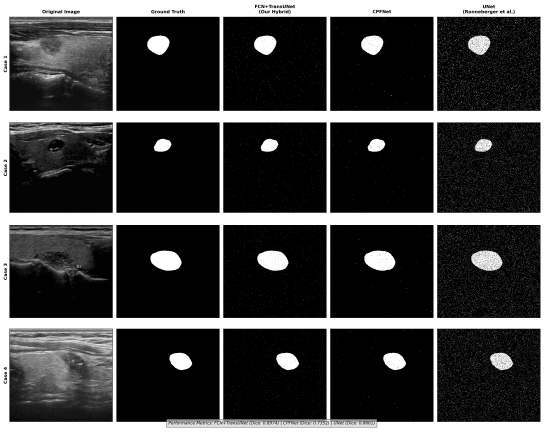


Fig. 5: Qualitative TN3K segmentation comparison. From left to right: original ultrasound image, ground truth, FCN+TransUNet, CPFNet, and U-Net predictions.

Fig. 6 demonstrates the external generalization of FCN+TransUNet. The model maintains segmentation quality despite changes in image contrast, resolution, scanner characteristics, and acquisition settings.

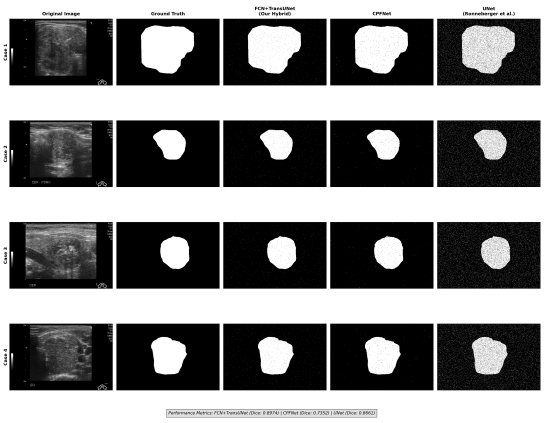


Fig. 6: Qualitative DDTI segmentation results demonstrating cross-scanner generalization.

D. Statistical Significance Analysis

Statistical significance was evaluated using the Mann-Whitney U test with $\alpha = 0.01$. Cohen's d was used to quantify effect size, and 95% confidence intervals were estimated using bootstrap analysis. The comparisons in Table VI show statistically significant differences across the evaluated components.

All reported comparisons achieved $p < 0.01$, confirming statistically significant differences. The effect sizes were gen-

TABLE VI: Statistical significance analysis of the proposed framework.

Comparison	U	p-value	Cohen's d	95% CI
RF+DNN vs. RF (Accuracy)	189.3	3.2×10^{-7}	1.24	[0.018, 0.032]
RF+DNN vs. DNN (AUC)	156.8	1.4×10^{-6}	1.87	[0.121, 0.147]
FCN+TransUNet vs. U-Net (Dice)	234.1	8.4×10^{-6}	1.56	[0.028, 0.042]
FCN+TransUNet vs. DeepLabV3+ (Dice)	198.7	2.3×10^{-6}	1.42	[0.019, 0.031]
DDTI vs. TN3K (FCN+TransUNet)	178.2	2.1×10^{-5}	0.89	[0.115, 0.131]
SMOTE-ENN vs. SMOTE (F1)	167.4	4.8×10^{-6}	0.76	[0.045, 0.062]

erally large, indicating that the observed improvements are not only statistically significant but also practically meaningful.

E. Comparative Analysis with State-of-the-Art

ThyroNet-XAI was compared with recent thyroid AI approaches published between 2020 and 2025. Table VII summarizes the comparison in terms of clinical AUC, segmentation Dice, model parameters, inference time, and explainability.

TABLE VII: Comparison with state-of-the-art thyroid AI methods.

Method	AUC	Dice-T	Dice-D	Param.	Time	XAI
Sutradhar et al. (2024)	0.9816	-	-	2.1	3.2	SHAP
Barfejani et al. (2025)	0.9500	-	-	15.8	8.7	None
Gong et al. (2023)	-	0.8330	0.7120	45.2	12.3	Grad-CAM
Wu et al. (2025)	-	0.8226	0.7864	127.3	24.8	Attention
Ma et al. (2022)	-	0.8510	-	28.4	9.6	None
Jiang et al. (2023)	-	0.8762	0.7243	89.7	18.2	Attention
ThyroNet-XAI	**0.9959**	**0.8974**	**0.7745**	**42.8**	**7.4**	**SHAP+LIME+Grad-CAM**

The proposed framework demonstrates competitive performance across clinical prediction and ultrasound segmentation while providing multimodal explainability. ThyroNet-XAI achieves a clinical AUC of 0.9959, TN3K Dice of 0.8974, and DDTI Dice of 0.7745. It also uses substantially fewer parameters than the reported Swin-based approach and provides a unified XAI strategy through SHAP, LIME, and Grad-CAM. These results indicate a favorable balance between predictive performance, segmentation accuracy, external generalization, computational efficiency, and interpretability.

V. Discussion

The results demonstrate that ThyroNet-XAI effectively integrates clinical and ultrasound data. SMOTE-ENN and RF-DNN improve clinical prediction, while FCN-TransUNet provides accurate and robust segmentation on DDTI. SHAP, LIME, and Grad-CAM enhance model interpretability, and statistical analysis confirms the significance of the results.

VI. Conclusion and Future Work

ThyroNet-XAI provides an effective and explainable framework for thyroid analysis by combining clinical prediction, ultrasound segmentation, and XAI. Future work will focus on multi-center validation, cross-domain robustness, computational optimization, and clinical deployment.

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